

Control Systems A

Course Introduction and Administrivia

UoN – FEE 3411

Sept – Dec 2026

- **FEE 3411 — Control Systems A** (45 notional hours)
- **Prerequisites:** FEE 222, FEE 372
- **Purpose:** introduce control engineering concepts and example control systems, and build analytical skills for time- and frequency-domain analysis of control systems
- **Delivery:** 2-hour lecture + 1-hour tutorial per week
- **Relation to FEE3412:** this course owns *analysis*; FEE3412 (Control Systems B, semester 2) owns *design* and revisits root locus, Bode and Nyquist as design tools

What you should be able to do by the end

- 1 Describe the working principles of a given control system
- 2 Analyse a control system solution in the time- and frequency-domains
- 3 Design a control system to control operations of a simple application (*introductory here; developed fully in FEE3412*)
- 4 Analyse the stability of a control system and design a strategy for its enhancement

Course content, in four blocks

- **Introduction** — open/closed-loop systems; components and devices (servos, stepper motors, hydraulics, pneumatics, actuators)
- **Time-domain analysis** — block diagrams, mathematical models, transfer functions, first- and second-order system response
- **S-domain analysis** — pole-zero maps, stability, Routh–Hurwitz, root locus, steady-state errors, an introduction to design and simulation
- **Frequency-domain analysis** — frequency response, Bode plots, Nyquist analysis and stability criterion, gain and phase margin, Nichols chart

How the semester is organised

- 15-week semester: Weeks 1–12 teaching, **Week 13 revision** (no new material), **Weeks 14–15** examinations
- Two CATs, each replacing that week's tutorial hour (the 2-hour lecture still runs):
 - **CAT 1** — Week 6 (covers Weeks 1–5)
 - **CAT 2** — Week 12 (covers Weeks 6–11)
- **Weeks 8, 10, 11** are hand-sketching tutorials (root locus, Bode, Nyquist) — come ready to draw, not just watch
- Week 9 includes a Simulink/python-control tutorial (gain-adjusted servo loop)

Assessment

Item	Timing	Weight	Coverage
Assignment 1	Issued Wk 1, due Wk 3	1.5%	Components; modelling
Assignment 2	Issued Wk 4, due Wk 6	1.5%	TFs, block diagrams, SFG
CAT 1	Week 6	12%	Weeks 1–5
Assignment 3	Issued Wk 7, due Wk 9	1.5%	Routh–Hurwitz, root locus
Assignment 4	Issued Wk 9, due Wk 11	1.5%	Simulation (homework)
CAT 2	Week 12	12%	Weeks 6–11
Main examination	Weeks 14–15	70%	Whole course

Bottom line

Continuous assessment 30%, main examination 70% — fixed by the syllabus.

Set texts

- **Primary:** Khalil, *Control Systems: An Introduction* — self-contained lecture notes will cover the material either way
- **Secondary:** Nise, *Control Systems Engineering* (6th ed.)
- **Also referenced:** Nagrath & Gopal, *Control Systems Engineering* (hardware chapter; contour mapping)
- **Further reading:** Sundararajan; DiStefano; Dorf & Bishop; Petkov et al.

Note

Lecture notes are written to be self-contained — you can learn from them without the textbook. Readings reinforce, they are not a prerequisite.

- **Venue / time:** *[to confirm]*
- **Office hours:** *[to confirm]*
- **Contact:** *[university email to confirm]*
- **Materials needed:** laptop or tablet for Week 9 (Simulink / python-control); otherwise pen and paper for hand-sketching weeks
- **Course materials** shared via *[platform to confirm — e.g. class WhatsApp / Google Classroom]*

Let's get started.